

Reliability-Aware Hybrid Retrieval for Natural-Language-to-Shell-Command Assistance: A Non-LLM Study

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Abstract

Natural-language interfaces to the shell are typically framed as a generation problem: translate an English request into a novel command (NL2Bash, NL2SH). We study a different, deliberately constrained point in the design space — closed-vocabulary *retrieval* from a curated, pre-vetted command library, using no large language model anywhere in the system. Starting from a real, previously-shipped BM25 baseline (67.3% overall accuracy on a validated 150-query benchmark, but with severe confidence miscalibration — 86.06% mean confidence on wrong answers), we test whether a lightweight hybrid lexical+local-semantic retrieval architecture, combined with post-hoc confidence calibration and margin-based rejection, can measurably improve both accuracy and reliability. We evaluate on two versions of a benchmark we constructed — an original 150-query version and an expanded 209-query version built specifically to probe statistical power on the original’s smaller subsets — under a fully leakage-free nested 5-fold cross-validation protocol. Hybrid fusion improves supported-task accuracy over pure BM25 on both benchmarks (71.9% $\rightarrow$ 77.1% and 75.4% $\rightarrow$ 79.3%), but under a pre-registered Holm–Bonferroni correction across a four-comparison primary family, this improvement **barely survives correction on the original benchmark (Holm-adjusted $p = 0.047$) and fails to survive on the expanded benchmark (raw $p = 0.070$)** — a non-replication we verified is not a computation error and report as a central finding. Isotonic-regression calibration reduces confidence miscalibration substantially and consistently (56.7–84.1% relative ECE reduction) and is the **only comparison that survives Holm correction on both benchmarks** (Holm-adjusted $p = 0.0004$, $p = 0.0003$). Out-of-domain rejection, initially underpowered on the original benchmark’s 15-query OOD subset (raw $p = 0.25$), is confirmed significant after correction on the expanded benchmark’s 50-query subset (Holm-

adjusted $p = 0.00006$). An intent-held-out (GroupKFold) generalization check shows the surviving accuracy and OOD findings are not artifacts of intent overlap between tuning and test, while calibration, though still substantial, is measurably attenuated on unseen intents (69–77% vs. 76–80% relative ECE reduction). A deterministic, rule-based safety classifier achieves 95%/95% precision/recall with zero dangerous-direction misses. An independent re-validation check on a stratified sample of the expanded benchmark’s AI-authored queries yields Cohen’s $\kappa = 0.63$ (below our $\kappa \geq 0.7$ target), with disagreement concentrated entirely in the ambiguous-query category and perfect agreement on out-of-domain queries — a finding we report as measured, not adjusted. We position this as the first reported study, to our knowledge, of a lexical/BM25 retrieval baseline for this task family evaluated with nested-CV calibration, OOD detection, a multiple-comparison-corrected benchmark-expansion replication check, and an intent-held-out generalization check — and we report every weaker, null, and non-replicating result alongside the positive ones.

1 Introduction

Natural-language shell assistance can lower the barrier to command-line tools, but incorrect commands — especially destructive ones — carry real cost. The dominant research trajectory since NL2Bash (Lin et al., 2018) has been generative: seq2seq and Transformer models (the NLC2CMD competition, Agarwal et al., 2021), then LLM-based systems (NL2SH; Westenfelder et al., 2025), most recently with execution-grounded, RL-trained approaches (BashCoder-R1; Yu et al., 2026). This trajectory inherits a persistent risk: open-vocabulary generation can hallucinate syntactically valid but semantically wrong or unsafe commands.

We study a different, narrower approach implemented in a real, previously-shipped, in-use npm package: retrieval from a fixed, pre-vetted corpus

of 431 records spanning 279 unique command intents on the evaluated platform, using classical lexical (BM25) scoring, with no LLM anywhere in the system. This constrains coverage but structurally eliminates hallucination of arbitrary commands. We independently audited and reproduced the system’s baseline behavior (§3) and found a specific, serious problem: the system is frequently *confidently* wrong, not merely wrong. This paper asks whether that problem, and the system’s accuracy more broadly, can be improved without abandoning the no-LLM, fully-offline design constraint.

Given a natural-language query and a fixed corpus of (intent, command, category, description, os) tuples, the task is to return the single best-matching command, or explicitly decline to answer if no corpus entry is a confident match — retrieval, not generation, over a closed and pre-vetted output space, unlike NL2Bash-style tasks where the model may emit any syntactically valid command string.

Research question. Can a lightweight hybrid lexical+local-semantic retrieval architecture, combined with explicit uncertainty estimation and post-hoc calibration, measurably improve accuracy and reliability over a pure-BM25 baseline, without the cost or hallucination risk of a generative system?

Contributions. (1) An independent, reproduced audit of a real shipped BM25 baseline’s accuracy and — more consequentially — its confidence miscalibration. (2) A five-component reliability-aware retrieval architecture evaluated under leakage-free nested cross-validation. (3) A replication check across two independently constructed benchmark versions, under a pre-registered Holm–Bonferroni correction, that overturns which of our own results we can actually claim are significant. (4) An intent-held-out (GroupKFold) generalization check. (5) Full, undiminished disclosure of every null, weak, or non-replicating result we found, including a below-target independent inter-annotator agreement check on our own benchmark’s AI-authored queries.

2 Related Work

NL2Bash (Lin et al., 2018) established the seq2seq/CopyNet/Tellina baseline and the BLEU/template/exact-match evaluation this field has used since. The NLC2CMD competition

(Agarwal et al., 2021) introduced a utility+flag-overlap metric more tolerant of near-miss answers than exact match. NL2SH (Westenfelder et al., 2025) is the modern generation-based successor, with even SOTA LLM accuracy reported as “remains low.” Recent work reinforces our framing that exact-match accuracy alone is an incomplete signal: QuoteBench (Li et al., 2026) shows matched execution scores can mask command-path failures, and BashCoder-R1 (Yu et al., 2026), despite RL training against execution feedback, tops out at 73–90% full functional success even as a strong system. A 2026 scaling study (Wang et al., 2026) finds BM25 overtakes dense and agentic retrieval past roughly 10M corpus tokens, supporting lexical retrieval’s competitiveness at scale, though at a corpus scale far larger than ours. Notaro et al. (2024) train a transformer-based command-risk classifier; we deliberately compare against a simpler, fully deterministic rule-based alternative (§7). No prior work, to our knowledge, reports a BM25 retrieval baseline for this task family evaluated under nested-CV calibration, OOD detection, and an intent-held-out generalization check — the gap this paper fills.

3 TermAssist: Baseline System and Benchmark

Baseline architecture (frozen for this study): tokenize (lowercase, strip punctuation, 10-word stopword list), BM25 scoring ($k_1 = 1.2$, $b = 0.75$, standard IDF smoothing) over intent+category+description, plus a flat +15.0 bonus when the query is a near-substring of the command’s intent. Confidence = $\min(\text{round}(\text{score}/8 \times 100), 100)$, rejected below score < 2.0. 431 corpus records, 27 categories, platform-filtered (win32/darwin/linux/all). No embeddings, no LLM, mean latency 3.29ms. We independently re-verified this baseline is byte-for-byte reproducible (0/150 mismatches against the archived results). **Corpus size clarification:** 134 of the 431 records are cross-platform duplicates of the same intent, so the retrieval space actually exercised on the evaluated (win32) platform is **279 unique intents, not 431** — the number we use throughout when comparing corpus scale to other work. We additively migrated the corpus schema (re-verified non-breaking) to add fields supporting a planned expansion to 500–800 human-validated intents; that expansion requires human authoring

and validation and is not yet complete, so all results here are reported on the original 279-intent corpus.

Benchmark, v0.1. 150 hand-adjudicated queries across 9 types (canonical, paraphrase, low-overlap-paraphrase, ambiguous, OOD, polysemy, single-keyword, safety-sensitive, complex-multi-intent), with ground-truth classification (119 CORRECT, 14 AMBIGUOUS, 15 OOD, 2 NEEDS_CORRECTION) and a risk-level field (120 LOW, 10 MEDIUM, 15 HIGH, 5 CRITICAL) reused in the safety evaluation (§7). We independently verified benchmark integrity: an apparent SHA-256 mismatch on Windows checkouts was root-caused to a CRLF/LF line-ending artifact, not content tampering.

Benchmark, v0.2. A second version (209 queries: the original 150 plus 35 new OOD and 24 new ambiguous queries) was built specifically to test whether v0.1’s small OOD (15) and ambiguous (14) subsets were statistically underpowered. The new queries were authored and adjudicated by an AI agent under human direction — disclosed explicitly — following the identical design/verification methodology as v0.1 (each query checked against actual system retrieval output, not judged from intuition; 6 of 30 drafted ambiguous candidates were rejected for failing the ambiguity or novelty criteria). We report results on both versions throughout, since §6 and §7 show they are not merely a larger sample of the same result.

Baseline results (v0.1). Overall accuracy 67.3% (101/150); supported-task (non-OOD) accuracy 71.9%; OOD rejection 26.7% (73.3% false-acceptance); ambiguity success 28.6%. By type: canonical/safety-sensitive 100%, paraphrase 88.0%, complex-multi-intent 86.7%, polysemy 55.6%, ambiguous 58.3%, low-overlap-paraphrase 33.3%, OOD 26.7%, single-keyword 0.0%. Mean confidence on the 49 wrong (non-rejected) predictions: **86.06%**, with 44.9% of failures at exactly 100% confidence — the central problem motivating this paper.

4 Reliability-Aware Retrieval Architecture

Five components, each targeting a specific measured failure: (1) local dense retrieval (Xenova/all-MiniLM-L6-v2, 384-dim, offline after one-time download) for the lexical gap; (2) hybrid score fusion with α tuned via nested

5-fold CV; (3) margin/entropy uncertainty signals over top-10 ranked candidates; (4) OOD and ambiguity detectors (dev-tuned thresholds); (5) isotonic confidence calibration. A sixth, independent component — a rule-based safety classifier — targets command risk, orthogonal to retrieval correctness. No LLM is used anywhere in the system.

5 Experimental Methodology

All tuned parameters (fusion α , OOD/ambiguity thresholds) are selected via nested 5-fold cross-validation (seed 42, stratified by classification) — **Split A:** for each test fold, the parameter is chosen using only the other four folds, then applied once to the held-out fold, never tuned on data it is later scored against. Statistical testing uses the exact (binomial) McNemar’s test, given small discordant-pair counts.

Multiple-comparison correction. We pre-register a primary family of four comparisons per benchmark version — accuracy vs. BM25, accuracy vs. dense-alone, OOD rejection, and calibration ECE reduction — and apply Holm–Bonferroni correction within each family; a result is called significant only at Holm-adjusted $p < 0.05$. The preregistration text and family definition predate the corrected results.

Bootstrap confidence intervals. Percentile bootstrap (10,000 resamples, seed 42) provides 95% CIs for the accuracy deltas (paired resampling over non-OOD query ids) and for the calibration ECE reduction (resampling over pooled held-out predictions), complementing the discrete exact McNemar’s test with a continuous-estimate view of the same comparisons.

Split B — intent-held-out generalization. To test whether Split A’s findings depend on the same command intents appearing in both tuning and test folds, we additionally construct a GroupKFold split keyed by a derived intent-group id (the gold command for answerable queries; the sorted acceptable-command set for ambiguous queries; a singleton per query for OOD queries, which target no intent). We run this on both benchmark versions (171 groups over v0.2’s 209 queries; 125 groups over v0.1’s 150 queries), both verified programmatically to place zero groups across more than one fold, so α , the detector thresholds, and the isotonic calibrator are always selected on queries

targeting *different* command intents than those they are scored on.

6 Results

Retrieval accuracy. On v0.1: BM25 71.9%, dense-only 72.6–72.7%, hybrid 77.1% (bootstrap 95% CI on the delta [2.2, 8.9]pp; exact McNemar’s $p = 0.0156$, **survives Holm correction, but only just:** Holm-adjusted $p = 0.047$, family size 4). On v0.2: BM25 75.4%, dense-only 72.3–72.4%, hybrid 79.3% (+3.8pp, bootstrap CI [0.6, 7.5]pp) — **this comparison does not clear $p < 0.05$ even before correction** (raw $p = 0.070$). At the per-query level, 7 of 8 discordant non-OOD pairs on v0.2 still favor hybrid, but one (TA-B194, “sudo reboot” vs. “sudo shutdown -h now”) now favors BM25, and the larger, harder query set is enough to push the p -value above the conventional threshold — not a computation error. The bootstrap CI on v0.2’s delta technically excludes zero while the discrete exact test does not reach significance; we treat this as **method-divergent, borderline evidence**, not a resolved significant result. Conversely, hybrid vs. dense-alone flips from not significant on v0.1 (Holm-adjusted $p = 0.292$) to significant on v0.2 even after correction (Holm-adjusted $p = 0.025$) — we attribute this, as an interpretation rather than a proven cause, to v0.2’s 17 new short-technical-keyword ambiguous queries (grep, sed, awk, tar, etc.), where BM25’s exact-token-match strength is plausibly more competitive; we did not further test this and flag it as future work.

By query type (v0.1), hybrid retains BM25’s 100% on canonical and safety-sensitive queries while improving most on low-overlap-paraphrase (Figure 2, left); single-keyword and OOD queries remain the hardest categories for both systems. The same breakdown on v0.2 (Figure 2, right) shows a similar overall shape with the new short-technical-keyword ambiguous queries added to that category.

OOD and ambiguity detection. On v0.1: AUROC 0.867 (OOD, feature = absolute top-1 score) and 0.784 (ambiguity, feature = margin) — an evidence-driven feature choice (§4), not selected post-hoc. On v0.2 (OOD subset expanded 15→50 queries): OOD AUROC improves to 0.901, and the baseline-vs-tuned rejection-rate improvement (34.0%→68.0%) **survives Holm correction** (Holm-adjusted $p = 0.00006$) — resolving the power limitation of the v0.1-scale result, which

BM	Comparison	Raw p	Holm	Sig.
v0.1	Acc., hybrid vs. BM25	0.0156	0.0469	Yes
v0.1	Acc., hybrid vs. dense	0.1460	0.2920	No
v0.1	OOD rej., tuned vs. base	0.25	0.2920	No
v0.1	Calib. ECE reduction	0.0001	0.0004	Yes
v0.2	Acc., hybrid vs. BM25	0.0703	0.0703	No
v0.2	Acc., hybrid vs. dense	0.0127	0.0255	Yes
v0.2	OOD rej., tuned vs. base	0.000015	0.00006	Yes
v0.2	Calib. ECE reduction	0.0001	0.0003	Yes

Table 1: Holm-corrected significance ($\alpha = 0.05$), both benchmarks. Bootstrap 95% CIs for each row are in §6 and the accompanying markdown source’s Table 1.

was not significant even before correction (raw $p = 0.25$). Ambiguity detection F1 improves on v0.2 (0.310→0.496) though AUROC slightly decreases (0.784→0.723), reported as measured.

Calibration. ECE is reduced 56.7–84.1% across four confidence variants tested, largest for the hybrid system’s own fused-score signal. A paired bootstrap test of the reduction gives one-sided $p \approx 0.0001$ on both benchmarks, and this is the **only comparison in the primary family that survives Holm correction on both benchmark versions** (Table 1). On v0.1, isotonic calibration reduces the hybrid system’s own reliability-diagram ECE from 0.2741 to 0.0537 (Figure 3, left); on v0.2 the same signal’s ECE reduces from 0.3237 to 0.0738 (Figure 3, right) — both large reductions, though the v0.2 starting ECE is itself higher, consistent with the harder query mix discussed throughout this section. Selective answering at 50% coverage achieves 10.7% error vs. 30.7% unconditional on v0.1 (Figure 4, left); the same protocol on v0.2 is shown in Figure 4 (right).

Intent-held-out generalization (Split B). GroupKFold over 171 derived intent groups (v0.2; 0 groups split across folds, verified programmatically) shows the two Holm-surviving v0.2 findings above are not artifacts of intent overlap between tuning and test: non-OOD accuracy is 79.24% under Split B vs. 79.25% under Split A, and OOD AUROC is 0.898 vs. 0.901 — both effectively unchanged. Calibration, while still a large improvement, is measurably **attenuated on held-out intents:** a 69.0% vs. 77.2% relative ECE reduction. The identical protocol run on v0.1 (125 groups) replicates the same qualitative pattern at smaller, noisier scale (Table 2) — consistent with the same v0.1-scale OOD underpowering already noted above, not a new artifact.

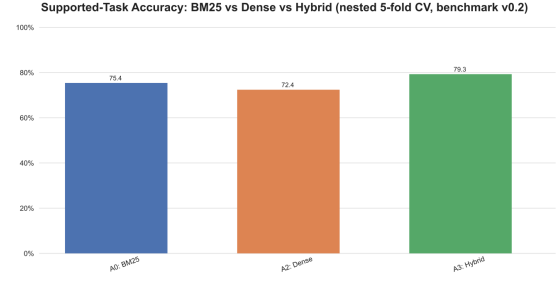
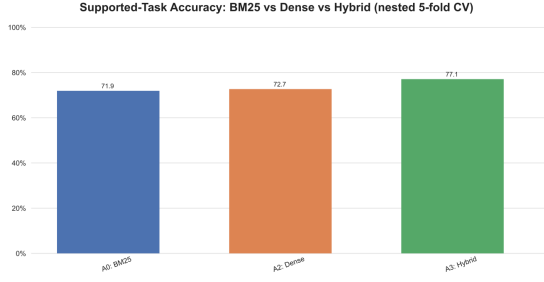


Figure 1: Supported-task accuracy for BM25, dense-only, and hybrid retrieval under nested 5-fold CV. Left: v0.1. Right: v0.2. Placed side by side to make the accuracy non-replication visually explicit – the bars look similar, but only the v0.2 dense-vs-hybrid comparison survives Holm correction (Table 1).

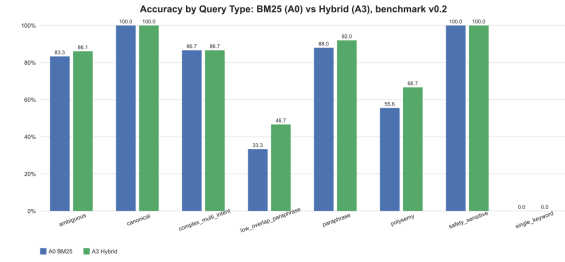
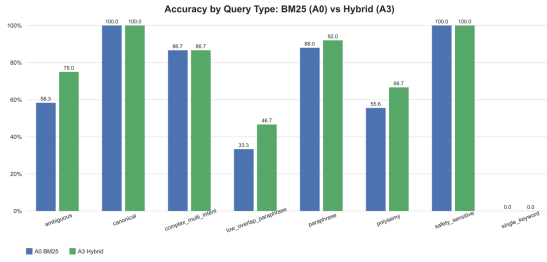


Figure 2: Accuracy by query type, BM25 (A0) vs. hybrid (A3). Left: v0.1. Right: v0.2.

BM	Metric	Split A	Split B	Δ
v0.1	Non-OOD acc.	77.06%	77.04%	-0.02pp
v0.1	OOD AUROC	0.867	0.849	-0.018
v0.1	Ambig. F1	0.310	0.300	-0.010
v0.1	Calib. ECE	.274→.054	.274→.065	+0.011
v0.2	Non-OOD acc.	79.25%	79.24%	-0.01pp
v0.2	OOD AUROC	0.901	0.898	-0.003
v0.2	Ambig. F1	0.496	0.479	-0.017
v0.2	Calib. ECE	.324→.074	.324→.101	+0.027

Table 2: Split A vs. Split B (intent-held-out generalization).

7 Ablation, Functional, and Safety Evaluation

Ablation (v0.1 → v0.2). A0 (BM25) = 71.9%/75.4%; A1 (BM25, no substring bonus) is **identical** to A0 on both benchmarks (0 command differences on 150 or 209 queries) — the bonus fires on 47/150 v0.1 queries but never changes the top-1 winner, a null result we report plainly rather than let the bonus’s presence imply an untested effect. A2 (dense) = 72.6–72.7%/72.4%; A3 (hybrid, nested-CV) = 77.0–77.1%/79.3%. A4 (+margin rejection) reaches 88.5% selective accuracy @ 70.8% coverage on v0.1; A5 (+OOD detection) reaches 90.1% @ 69.5% coverage (10/15 OOD caught). On v0.2, A5 is *lower* at lower coverage (82.2% @ 58.9%, 38/50 OOD

caught) — the harder v0.2 query mix making the accept/reject decision more difficult, not a regression in the method. A6 (+calibration) makes the same decisions as A5 but with ECE 0.054 vs. 0.274 raw.

Functional evaluation. Sandboxed execution (disposable temp directories, never the host machine) on a deliberately narrow, explicitly-scoped 15/150-query subset (git + filesystem, placeholder-free, non-interactive, host-safe gold commands only). Gold commands: 100% functional success (validates benchmark quality). Retrieved (hybrid) commands: 93.3% functional success, zero cases of textually-correct-but-functionally-broken — a positive result that covers only 10% of the benchmark by design.

Safety evaluation. A fully deterministic, rule-based classifier (LOW/MEDIUM/HIGH/CRITICAL), authored from general domain knowledge before inspecting per-query outcomes, evaluated against the benchmark’s pre-existing risk-level ground truth: 89.6% exact 4-tier accuracy (112/125 gold commands), 95%/95% precision/recall on the actionable risky-vs-not-risky binary, zero dangerous-direction misses (no CRITICAL/HIGH command ever tagged safe). On retrieved commands, accuracy

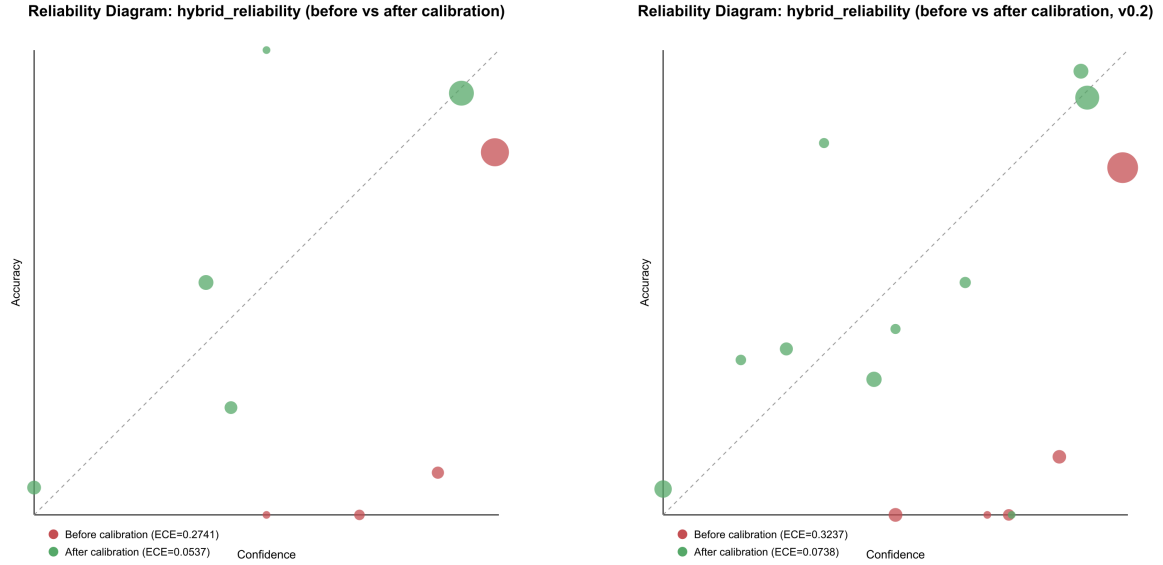


Figure 3: Reliability diagram for the hybrid system’s fused-score confidence, before vs. after isotonic calibration. Left: v0.1. Right: v0.2.

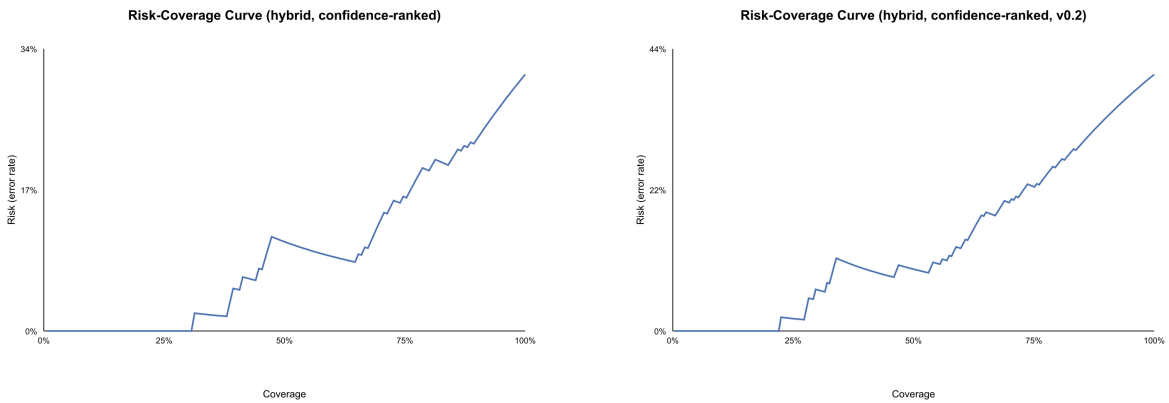


Figure 4: Risk-coverage curve for the hybrid system under confidence-ranked selective answering. Left: v0.1. Right: v0.2.

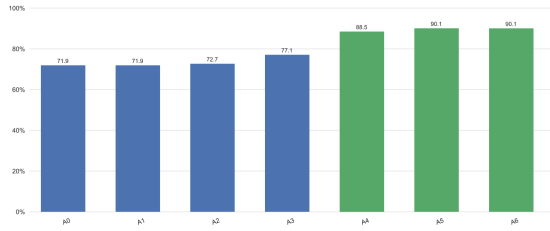
holds (90.0%) but risky-binary precision drops to 0.826 — a retrieval-error artifact, not a classifier defect. On v0.2, exact accuracy and recall shift modestly downward (83.9%/0.864) because the classifier — deliberately not re-tuned against new examples — was authored before v0.2’s new gold commands existed, surfacing the same documented gaps, not new ones.

8 Independent Re-Validation of AI-Authored Benchmark Queries

v0.2’s 59 new queries were authored and adjudicated by an AI agent under human direction. To check this did not silently introduce systematic labeling error, we drew a stratified, blind, seed-42 sample of 14 of these 59 queries (8 OOD, 6

ambiguous) and had an independent reviewer — blind to the original label of each specific query — judge them fresh into one of three categories. Result: **Cohen’s $\kappa = 0.6316$** , below our pre-specified $\kappa \geq 0.7$ target, reported as measured. The disagreement is not uniform: agreement was **perfect on the 8 OOD-labeled items (8/8)** and entirely concentrated in the **6 ambiguous-labeled items (3/6)**. All three disagreements were short, bare-keyword-style queries (“list running processes”, “ffmpeg”, “pip”) that our labeling called ambiguous — following v0.1’s own precedent of treating bare tool-name keywords like `git/docker` as ambiguous — while the independent reviewer judged each to have one sufficiently obvious default command. We read this as corroborating, not weakening, the OOD-significance claim (§6, Ta-

lation A0-A6 (A0-A3: unconditional accuracy; A4-A6: selective accuracy at reduced coverage -- not directly comparab



on A0-A6 (A0-A3: unconditional accuracy; A4-A6: selective accuracy at reduced coverage -- not directly comparable)

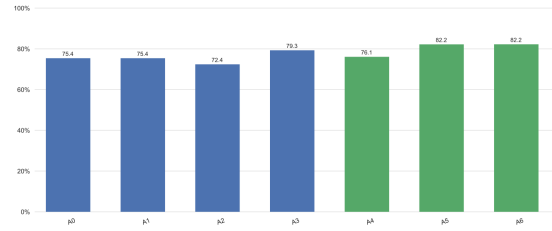


Figure 5: Ablation A0–A6. A0–A3 report unconditional accuracy; A4–A6 report selective accuracy at reduced coverage and are not directly comparable to A0–A3’s bars. Left: v0.1. Right: v0.2, visualizing A5’s lower selective accuracy at lower coverage discussed above.

ble 1) — which rests on the OOD label set, independently validated here — while sharpening the already-disclosed ambiguity-detection weakness with a specific, localized cause. We also disclose that reviewer independence here is *partial*: the reviewer directed the original authoring process at a summary level, though was blind to which label each of these 14 specific queries had received. A fully independent, non-project-affiliated annotator reaching $\kappa \geq 0.7$ remains open future work.

9 Discussion

The central finding is nuanced, not a clean win, and we report it that way. Hybrid retrieval measurably improves both *what* the system gets right and *how honestly* it communicates uncertainty about what it gets wrong — two related but distinct problems, requiring two different interventions, neither of which alone was sufficient (84.8% of the hybrid system’s remaining errors on v0.1 are still high-confidence, improving only modestly to 67.5% on v0.2). But the accuracy improvement’s statistical significance is benchmark-composition-sensitive: it barely holds on the original 150-query benchmark and does not replicate on the expanded, harder 209-query version, while a different comparison (hybrid vs. dense-alone) becomes significant, and survives correction, only on the expanded version. Applying Holm–Bonferroni correction sharpens this picture rather than changing it qualitatively: of the four pre-registered comparisons, only calibration ECE reduction survives correction on both benchmarks, OOD detection survives correction only on the powered v0.2 benchmark, and the BM25-vs-hybrid accuracy comparison survives correction only marginally on v0.1 and not at all on v0.2. We interpret this as evidence that a single benchmark, however carefully constructed, can

support a significance claim that is real but fragile to the specific mix of query difficulty sampled — precisely the concern Card et al. (2020) raise about NLP benchmark power, and precisely why we report both benchmark versions’ results, under correction, rather than only the more favorable one. The intent-held-out check adds an independent line of evidence for the same conclusion: the two findings that survive correction also generalize essentially unchanged to withheld command intents, while calibration — still a robust improvement — is measurably weaker on those same held-out intents. The independent κ check (§8) adds a third, orthogonal line of evidence pointing the same direction: the OOD claim is corroborated from yet another angle, while the weakest claim (ambiguity) accumulates yet another disclosed weakness. Together, this argues that reliability, not raw accuracy, is this study’s most defensible individual contribution — not an artifact of any single benchmark’s composition, of intent overlap between tuning and test, or of label-authoring method.

Limitations

- **Benchmark composition sensitivity.** The full evaluation suite was re-run on two independently constructed benchmark versions and all primary comparisons were re-evaluated under Holm correction. The OOD-significance gap on v0.1’s 15-query subset was resolved on v0.2’s 50-query expansion, but the core BM25-vs-hybrid accuracy significance did **not** survive correction robustly on either benchmark (marginal on v0.1, absent on v0.2) — neither benchmark version alone should be treated as definitive.
- **Canonical-query independence.** All 25 canonical queries are verbatim-equal to a cor-

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pus intent, so 100% canonical accuracy is a positive-control sanity check, not evidence of language-understanding capability; the paraphrase, low-overlap, polysemy, and ambiguous types carry the actual capability claims.

- **Corpus scale.** 279 win32-visible unique intents (431 raw records), small relative to generation-oriented corpora; an additive schema migration is complete, but the planned 500–800 intent expansion is not, pending human authoring and validation. Results are single-platform (win32).
- **Query authorship.** v0.2’s 59 new queries were authored and adjudicated by an AI agent under human direction, disclosed explicitly. Independent re-validation (§8) on a stratified sample yielded $\kappa = 0.63$, below our $\kappa \geq 0.7$ target, with disagreement concentrated in the ambiguous category and perfect agreement on OOD queries; reviewer independence was partial, not a fully uninvolved third party.
- **No large-scale human evaluation or preference study,** fixed embedding-model choice, and narrow (10%) functional-evaluation coverage (confirmed unchanged between benchmark versions without re-executing identical sandboxed commands).
- **Ambiguity detection remains an unresolved weakness** (F1 0.48–0.50 under both class-stratified and intent-held-out evaluation), now further explained by §8’s finding that bare-keyword queries are specifically where independent annotators disagree with our own ambiguous labels.
- **Isotonic calibration’s sensitivity to small samples and intent generalization** is measured directly, not merely hypothesized: the ECE reduction attenuates from 77% within-distribution to 69% on intent-held-out evaluation.
- **Construct validity.** “Functional success” is defined as exit-code-0, not full semantic side-effect verification.
- **External validity.** Results are specific to this 279-intent, win32-only corpus and may not generalize to larger or differently distributed command libraries; the intent-held-out check

addresses generalization *within* this corpus’s intent distribution, not beyond it.

Ethics Statement

TermAssist retrieves from a fixed, human-curated command library rather than generating arbitrary shell commands, which structurally bounds the space of possible harmful outputs to commands a human already vetted — but a retrieved command can still be inappropriate for a given user’s intent, and our own baseline audit found the deployed system is frequently *confidently* wrong. We evaluate command risk with a separate, deterministic safety classifier (§7) specifically because retrieval correctness and command risk are orthogonal concerns; a correctly-retrieved command can still be destructive if misapplied, and no component in this study executes a retrieved command against a user’s real system. All functional-evaluation command execution in this study ran in disposable, sandboxed temporary directories, never against a host machine. We disclose the AI-assisted authorship of part of our benchmark (§3) and the below-target result of our own attempt to independently validate it (§8) rather than omit or minimize either.

10 Conclusion and Future Work

A lightweight, fully offline, non-LLM hybrid retrieval architecture improves accuracy and substantially improves confidence calibration over a real, previously-shipped BM25 baseline, evaluated under a leakage-free nested cross-validation protocol, re-run in full on two independently constructed benchmark versions, and further validated under a pre-registered Holm–Bonferroni correction, an intent-held-out generalization check, and an independent inter-annotator agreement check. The most defensible, replication-tested, multiple-comparison-corrected, and generalization-checked claims from this program are the calibration improvement and the OOD detection improvement — not the raw accuracy gain in isolation, which does not replicate robustly across benchmark versions. Open future work includes: a direct investigation of why v0.2’s short-technical-keyword ambiguous queries shift the accuracy-significance pattern (§6; we offer an interpretation, not a tested cause); a richer ambiguity-detection feature, since a larger ambiguous subset improved F1 but not AU-ROC under either split; expanding the corpus from 279 to 500–800 human-validated intents; reaching $\kappa \geq 0.7$ on the independent query-validation

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check with a fully independent annotator; an independent, human-labeled safety-evaluation set not derived from the rule-based classifier’s own logic; expanding functional evaluation from 15 to roughly 40–60 sandboxed tasks; and a human-preference study. None of these can be responsibly completed by bulk automated generation or by the same process that produced the system under test, which is precisely why they remain future work rather than attempted here.

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